

downwards to touch the horizontal axis.

- 29** An atom makes a transition from an energy level of energy  $E_1$  to a level of energy  $E_2$ , emitting a photon of wavelength  $\lambda$ .

Which expression gives this wavelength in terms of Planck constant  $h$  and the speed of light  $c$ ?

**A**  $\frac{E_1}{hc} - \frac{E_2}{hc}$

**B**  $\frac{hc}{E_1} - \frac{hc}{E_2}$

**C**  $\frac{hc}{E_2} - \frac{hc}{E_1}$

**D**  $\frac{hc}{E_1 - E_2}$

Ans: D

$$E_1 - E_2 = \frac{hc}{\lambda}$$

$$\lambda = \frac{hc}{E_1 - E_2}$$

**30** A proton and an alpha particle ( ${}^4\text{He}$  Helium nucleus) have the same de Broglie wavelength